

MEC - 107 INTERNATIONAL TRADE & DEVELOPMENT



INTERNATIONAL TRADE AND DEVELOPMENT

(Vol. 1)



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BLOCK 2	MODERN THEORIES OF INTERNATIONAL TRADE	
Unit 3	Intra-Industry Trade	Prof. C Veeramani, Indira Gandhi Institute of Development Research, Mumbai
Unit 4	Alternative Explanations of Trade	Dr. Pravin Jadhav, Assistant Professor, Institute of Infrastructure Technology Research and Management (IITRAM)
BLOCK 3	FREE TRADE VERSUS PROTECTIONISM	
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Unit 6	Instruments of Protectionism	Adapted from Unit 4 of the course MEC 004 written by Dr. Ananya Ghosh Dastidar
BLOCK 4	EXCHANGE RATE AND BALANCE OF PAYMENTS	
Unit 7	Exchange Rate Regimes	Prof. S.K. Singh, Retd. Prof. IGNOU
Unit 8	Components of Balance of Payments	
Unit 9	Impossible Trinity: Alternative Scenarios	
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Unit 14	Regional Trading Agreements	
Unit 15	India and Multilateral Trading System	
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Unit 17	Trade and Environment	Prof. S.N. Mukherjee, National Institute of Public Finance and Policy
BLOCK 8	INDIA'S FOREIGN TRADE: POLICY AND CHALLENGES	
Unit 18	India's Trade Policy	Ashish Gupta, Motilal Nehru College, Delhi University
Unit 19	India's Trade: Trends, Composition and Challenges	Dr. Pravin Jadhav, Assistant Professor, Institute of Infrastructure Technology Research and Management (IITRAM)



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COURSE INTRODUCTION

The global economy is constantly evolving, shaped by technological advancements, geopolitical shifts, and changing trade policies. The process of globalization has interconnected economies around the world. Increased cross-border trade, investment, and the international mobility of capital, labour, and technology have facilitated the integration of national economies into the global marketplace. Further, geopolitical dynamics and power shifts among nations have profound effects on cross-border flows and global economic integration. In light of the ever-changing, dynamic nature of the global economy, the course **MEC 107 'International Trade and Development'** aims to understand these drivers of international (both inter and intra-industry) trade and international finance.

This course will provide you with valuable knowledge and skills to understand and navigate the complexities of the global economy. International trade plays a crucial role in the economic development of nations. By studying this course, learners will gain a deep understanding of the factors that influence global trade patterns, such as trade policies, tariffs, exchange rates, and trade agreements. Geopolitical factors play a crucial role in shaping the global economy. Political stability, regional conflicts, and international relations impact trade flows, investment patterns, and economic alliances. Studying international trade and finance will equip the learners with the knowledge to analyze geopolitical developments and their implications on international business.

The course will also throw light into the debate of free trade versus protectionism by discussing what happens when countries adopt various instruments of protectionism and how the world is affected. In the light of changing nature of agreements between the countries and the new world order, the course also delves into various types of trade agreements and India's agreements with various nations. It also digs deeper into the evolved role of India in WTO and traces the evolving role since the GATT era. India's foreign trade policy, how it has evolved and what are the challenges associated with it are also discussed in depth. The course will equip the learners with an understanding of the impact of various policies of countries and international organisations that affect global patterns of trade and finance.

The course is structured around eight blocks. Each block is further subdivided into several units. Each Unit is self-contained and has linkages with all other Units.

Block 1: Theory of International Trade introduces the course by discussing the rationale behind International trade and why countries engage in trade. This block examines various theories of International trade and presents their comparative analysis. It identifies the various gains associated with international trade and how they can be measured. It also presents the comparative analysis of various types of gains in this block. As we move forward, we discuss the complementary theories of international trade that are formed by dropping the classical assumptions.

In **Block 2: Modern Theories of International Trade**. This block also delves into intra industry trade (IIT) i.e. the exchange of similar or closely related goods and services within the same

industry between countries or regions. IIT arises due to product differentiation and the diversity of consumer preferences is often driven by economies of scale and the ability to achieve cost efficiencies through mass production. We discuss the need, rationale and measurement of Intra-Industry Trade (IIT) in this block.

After discussing the need and theories of inter-industry and intra-industry trade, we present the two contrasting approaches to international trade: Free trade and protectionism, representing different policy stances and perspectives on the regulation of trade flows in **Block 3: Free trade versus protectionism**. We discuss the latest developments in India's Protectionism Policy and their implications for the Indian economy. To understand how countries adopt free trade and protectionist policies based on their specific economic and political considerations, we also discuss various instruments of protectionism-tariff and non-tariff barriers in this block. We examine their impact on the world economy as well.

As we move to **Block 4: Exchange Rate and Balance of Payments**, we examine the vital role that exchange rates and the Balance of Payments play in a country's macroeconomic stability. Exchange rate policies have significant implications for a country's trade competitiveness, economic growth, and financial stability. Some countries adopt fixed exchange rate systems, while others have floating or managed exchange rate regimes. In this block, we study exchange rates and their management to comprehend the advantages, disadvantages, and challenges associated with different exchange rate systems. We also trace the historical development of international exchange rate regimes and examine operational aspects of the current exchange rate regimes. We discuss the balance of payment disequilibrium and the various approaches that countries adopt for BOP adjustment. We also analyse the experience of various countries to understand the Macroeconomic policy trilemma, the impossible trinity.

International finance is closely intertwined with international trade. **Block 5: International Finance** will help to gain an understanding of global financial markets, capital flows, and international investment. Block 5 gives an idea of the globalization of financial markets and then the concept of International financial markets. It throws light on the global financial crisis of the year 2008. To understand this crisis better, some of the most devastating financial crises in history are also presented which will add to the understanding of the unfolding of the 2008 crisis. The block also summarises initial monetary and fiscal policy responses to manage this global crisis.

Since Globalization has opened immense opportunities for businesses to expand beyond national borders, we deal in depth with globalization in **Block 6: Globalisation**. The role of the World Trade Organization (WTO) and other multilateral organizations is becoming increasingly important as the world has become more interconnected than ever before, with increased cross-border trade, investment, and economic integration. This block furnishes an understanding of the history and present status of the multilateral trading system in terms of GATT, and WTO rounds. We also discuss the types of regional trading agreements, preferential trading arrangements, free trade areas and customs unions. India's role has also evolved in the multilateral trading systems and the agreements of which India is an integral part e.g., the Asia-Pacific Trade agreement,

ASEAN-India free trade area, SAFTA. The block also presents the contribution of India to WTO and how India has transformed itself with the change of GATT towards the WTO regime.

In **Block 7: Trade, Growth and Development**, we describe the relationship between economic growth and trade. The block presents the various trade policy regimes vis-à-vis growth in developing countries wherein the import substitution and outward-oriented development strategies are discussed. The block also discusses India's experience with inward-oriented and outward-oriented regimes. Because of the increased concern about climate change, we also discuss the link of trade with climate change and the policies and multilateral environmental agreements concerning trade.

Block 8: India's Foreign Trade: Policy and Challenges digs into the concepts, nature and aims of trade policy and the basic tools of a trade policy. It will further explain the evolution of the trade policy in India. It also presents the pattern, direction and composition of India's foreign trade. The challenges faced by the foreign trade of India be highlighted in the concluding part of the block.



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BLOCK 1

THEORY OF INTERNATIONAL TRADE

Unit 1	Classical and Neo-Classical Theories of International Trade
Unit 2	Gains from Trade

BLOCK INTRODUCTION

This block reveals the rationale behind International trade and why countries engage in trade. It presents a comparative picture of various theories of International trade. It highlights the various gains associated with international trade and how they can be measured. A comparative picture of various types of gains has also been presented in this block.

This block consists of 2 units.

Unit 1: Classical and Neo-Classical Theories of International Trade discusses the major theories of international trade. It presents the rationale behind why countries trade with each other. The unit covers discussion on major theories, like absolute advantage theory, comparative theory, and H-O theory. Further, the theories like Stolper – Samuelson theorem, Factor Price Equalisation theory, and Rybczynski theorem developed by extending the H-O model have also been discussed.

Unit 2: Gains from International Trade throws light on sources of gains, factors determining gains from trade and how these gains can be measured. It also discusses the concept of the Production Possibilities Curve in International trade. The unit also elucidates the situation of the potential and actual gain, Free Trade versus no trade and static and dynamic gains.

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UNIT 1 CLASSICAL AND NEO-CLASSICAL THEORIES OF INTERNATIONAL TRADE*

Structure

- 1.0 Objectives
- 1.1 Introduction
- 1.2 Theory of Mercantilism
- 1.3 Absolute Advantage Theory
- 1.4 Comparative Advantage Theory
- 1.5 Heckscher–Ohlin Theory
- 1.6 Stolper – Samuelson Theorem
- 1.7 Factor-Price Equalization Theorem
- 1.8 Rybczynski Theorem
- 1.9 Lets Sum Up
- 1.10 Key Words
- 1.11 Some useful References
- 1.12 Answers/Hints to Check Your Progress Exercise

1.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explain the difference between Domestic and International Trade;
- describe why two nations trade with each other;
- explain Absolute Advantage Theory;
- explain Comparative Advantage Theory;
- differentiate between Absolute Advantage Theory and Comparative Advantage Theory;
- describe Heckscher–Ohlin Theory;
- explain the Stolper – Samuelson theorem;
- explain Factor Price Equalisation theory; and
- describe Rybczynski Theorem.

* Contributed by Sh. Rahul Chaudhary, Consultant IGNOU.

1.1 INTRODUCTION

In this unit, we will discuss the major theories of international trade. At the beginning of this unit, we will try to understand what international trade is and how it differs from domestic trade. The reasons why countries trade with each other will also be discussed. Having understood these, the major theories of international trade that have evolved, like absolute advantage theory, comparative theory, and H-O theory, will be explained. Further, the theories like Stolper – Samuelson theorem, Factor Price Equalisation theory, and Rybczynski theorem developed by extending the H-O model will also be discussed.

Before reading the theories of international trade you should understand what is, international trade, and why does trade take place?

In simple words, trade refers to buying and selling goods and services in exchange for money. The sellers sell the goods and services while the buyers buy the same. When these activities occur domestically in the country, it is referred to as domestic trade or only trade. But, when the exchange of goods and services takes place between buyer and seller of two different nations it is called international trade.

Now, the question is, why do we trade at all?

The simple answer to this question is that any nation cannot produce everything it needs. One nation may have plenty of natural resources but a scarcity of meat, fish, etc. One country may be an efficient milk producer but have a scarcity of wheat and rice. In this situation, nations come together and exchange goods among them. We will discuss more about gains from trade in Unit 2.

1.2 THEORY OF MERCANTILISM

Mercantilism is an exercise that is more than 500 years old. The base of this theory was the "commercial revolution", the transition from local economies to national economies, from feudalism to capitalism, and from a rudimentary trade to a larger international trade.

The theory of mercantilism postulates that countries should encourage exports and discourage imports. The tendency to export more and import less and receive gold (as gold was the medium of exchange) in exchange is called **Mercantilism**. Mercantilism was the economic system of the major trading nations during the 16th to 18th centuries. The theory assumed that national wealth and power were best served by increasing exports and collecting precious metals like gold in return. The theory states that government should play a vital role in regulating the economy to encourage exports and discourage imports by using subsidies and taxes. According to this theory, the government should accumulate as much gold as possible, which can only be done through exports.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Explain the difference between Domestic and International Trade.

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2) Write down a short note on Mercantilism.

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1.3 ABSOLUTE ADVANTAGE THEORY

Several theories have been developed in light of the activities discussed above. These theories have evolved for more than 200 years now. Adam Smith propounded the basic theory of international trade in 1776. Adam Smith in his seminal work 'An inquiry into the Nature and Causes of the Wealth of the Nation' for the first time, provided a theoretical explanation of why trade should take place between two nations.

Adam Smith, in his book propounded the theory of absolute advantage, which states that a country should specialize in those products which it can produce efficiently, where efficiency is measured in terms of absolute labour costs. This theory assumes that there is only one factor of production: labour.

Adam Smith wrote in 'The Wealth of Nations', "If a foreign country can supply us with a commodity cheaper than we can make it, better buy it of them with some part of the produce of our industry, employed in a way in which we have some advantage".

According to Adam Smith, trade between two nations is based on absolute cost advantage. When one is more efficient than (or has an absolute advantage over) another in the production of one commodity but is less efficient than (or has an absolute disadvantage with respect to) the other nation in producing a second commodity, then both the nations can gain by each specializing in the production of the commodity of its absolute advantage and exchanging part of its output with the other nation for the commodity of its absolute disadvantage. In this process, the resources are utilized most efficiently and the output of both commodities will rise. This increase in output of both the commodities measures the Gains from specialization in production available to be divided between the two nations through trade.

In very simple words, he stated that trade would benefit both countries if country A exported the goods, which it can produce at a lower cost than country B and imported the goods, which country B can produce at a lower cost than it.

The above description of the theory can be understood better with the following example. Suppose there are two countries A and B, which produce wheat and rice with an equal amount of resources i.e. 200 labourers. Country A uses 10 labourers to produce 1 ton of wheat and 20 labourers to produce 1 ton of rice. Country B uses 25 units of labourers to produce 1 ton of wheat and 5 units of labourers to produce 1 ton of rice.

Table 1.1 Number of labourers used by Country A and Country B for producing 1 ton of Wheat and Rice

	Country A	Country B
Wheat (1 ton)	10 labourers	25 labourers
Rice (1 ton)	20 labourers	5 labourers

This is evident from Table 1.1 that country A has an absolute advantage in producing wheat as it can produce 1 ton of wheat by using fewer labourers as compared to country B. On the other hand, country B has an absolute advantage in producing rice as it can produce 1 ton of rice by employing fewer labourers in comparison to country A. Now, if there is no trade between these countries and resources (in this case there are a total of 200 labourers) are being used equally to produce wheat and rice, country A would produce 10 tons of wheat ($200/10$), and 5 tons of rice ($200/20$) and country B would produce 4 tons of wheat ($200/25$) and 20 tons of rice ($200/5$). Thus, total production without trade (i.e., autarky) is 39 tons (14 tons of wheat and 25 tons of rice).

Table 1.2 Production volume of wheat and rice without trade

	Country A (in Tons)	Country B (in Tons)
Wheat	10	4
Rice	5	20

Now, if both countries open up to trade with each other and specialize in goods in which they have an absolute advantage, the total production would be higher. If a trade takes place, Country A would produce 20 tons of wheat with 200 labourers; whereas, country B would produce 40 tons of rice with 200 labourers. Thus, total production would be 60 units (20 tons of wheat and 40 tons of rice).

Table 1.3 Production volume of wheat and rice with trade

	Country A (in Tons)	Country B (in Tons)
Wheat	20	0
Rice	0	40

It is clear from the above descriptive example that without specialization, the total production of countries was 39 tons which increased to 60 tons after specialization from trade opportunities. This is the welfare gains from trade. Therefore, the theory of absolute advantages shows that pattern of trade based on absolute cost differences would benefit both the countries.

1.4 COMPARATIVE ADVANTAGE THEORY

Elaborating on the theory of absolute advantage, David Ricardo presented his work in 1817 in his book 'Principles of Political Economy and Taxation'. In this book, he presented the law of comparative advantage. This is one of the most important and still unchallenged laws of economics with many practical applications. The Ricardian theory states that trade can be beneficial for two countries even if one country has an absolute advantage in all the products and the other country has no absolute advantage in any of the products. Ricardo argued that “...a nation, like a person, gains from the trade by exporting the goods or services in which it has its greatest comparative advantage in productivity and importing those in which it has the least comparative advantage”. A nation behaves no differently from an individual who does not attempt to produce all the commodities he needs. Rather, he produces only that commodity that he can produce most efficiently and then exchanges part of his output for the other commodities he needs or wants. The first nation would be better off provided they specialize in the production and export of those commodities in which its absolute disadvantage is smaller (this is the commodity of its comparative advantage) and import the commodity in which its absolute disadvantage is greater (this is the commodity of its comparative disadvantage).

This theory assumes that labour is the only factor of production in two countries, with zero transport cost, and no trade barriers within the countries. This theory can be understood better from the example cited in Table 1.4.

Suppose there are two countries, A and B, producing two commodities, wheat and rice, with labour as the only factor of production. Now assume that both the countries have 200 labourers and use 100 labourers to produce wheat and 100 labourers to produce rice.

Table 1.4 Production of wheat and rice by Country A and B before the trade

	Country A (in tons)	Country B (in tons)
Wheat	20	15
Rice	40	10

Table 1.4 shows that country A can produce 20 units; whereas, country B can produce 15 units of wheat by employing the same number (100) of labourers. In addition, country A can produce 40 units; whereas country B can produce 10 units of rice by employing 100 labourers.

Thus, country A has an absolute advantage in producing both items. Country A employs the same number of labourers (100 labourers in the production of each item) in producing rice and wheat; however, rice production is higher than wheat production. This reveals that country A has a comparative advantage in producing rice. Similarly, country B also employs the same number of labourers (100 labourers in producing each good) in manufacturing wheat and rice; but, its wheat production is more than rice. It indicates that country B has a comparative advantage in producing wheat.

For example, country A has decided to produce 60 units of rice by employing 150 labourers. It uses 50 labourers to produce 10 units of wheat. On the other hand, country B has decided to use all the 200 labourers to produce 30 units of wheat and stop the production of rice. In this situation, country A exchanges 14 units of rice with 14 units of wheat produced by country B. (see Table 1.5).

Table 1.5 Production after specialization

	Country A	Country B
Wheat	10	30
Rice	60	0

Table 1.6 Situation after the trade takes place

	Country A	Country B
Wheat	24	16
Rice	46	14

It can be understood from Table 1.6 that both countries have benefitted or gained from trade. Before the trade, country A had only 20 units of wheat and 40 units of rice; after the trade, country A had 24 units of wheat and 46 units of rice.

At the same time, country B has 15 units of wheat and 10 units of rice before the trade, while it has 16 units of wheat and 14 units of rice after the trade. Therefore, comparative advantage theory clarifies that trade can create benefits for both participating countries even if one country has an absolute advantage in the production of both commodities.

Thus, it is clear from the above theories that the mercantilists believed that one nation could gain only at the expense of another nation and advocated strict government control and regulations of all economic activities. Adam Smith believed that all nations would gain from free trade and strongly advocated a policy of laissez-faire (i.e., as little government interference with the economic system as possible). Free trade would cause world resources to be utilized most efficiently and would maximize world welfare. There were to be only a few exceptions to this policy of laissez-faire and free trade. One of these was the protection of industries important for national defence.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the central proposition of the absolute advantage theory and comparative advantage theory.

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- 2) Differentiate between Adam Smith and Ricardo's theory of international trade.

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- 3) With reference to the Ricardian theory of international trade, explain how nations gain from trade.

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1.5 HECKSCHER -OHLIN THEORY

In the preceding section, we have seen that the difference in relative commodity prices between two nations is the reason for their comparative advantage, which forms the basis for mutually beneficial trade. Then we explained the reason, or cause, for the difference in relative commodity prices and comparative advantage between the two nations. The second way we extend our trade model is to analyze the effect of international trade on the earnings of factors of production in the two trading nations. That is, we want to examine the effect of international trade on the earnings of labour as well as on international differences in earnings. These two important questions were left largely unanswered by Smith, Ricardo, and Mill. According to classical economists, the comparative advantage was based on the difference in the productivity of labour among nations, but they did not explain such a difference in productivity, except for possible differences in climate. The Heckscher–Ohlin theory goes much beyond that by extending the trade model of the previous sections to examine the basis for comparative advantage and the effect that trade has on factor earnings in the two nations. (Salvatore 2013)

Bertil Ohlin has advocated the Modern Theory of international trade. Ohlin has drawn his ideas from Eli Heckscher's General Equilibrium Analysis. Hence it is also known as Heckscher Ohlin (H-O) Theory.

Heckscher-Ohlin theory begins where the Ricardian theory of international trade ends. The Ricardian theory states that the basis of international trade is the comparative or relative cost differences. But he did not explain how this comparative cost difference emerges. Ohlin's theory explains the real cause of this difference.

Ohlin states that trade takes place due to the different relative prices of different goods in different countries. The difference in the relative price of a commodity is due to differences in the relative costs and factor prices in different countries.

According to the H-O theory, the differences in factor prices are due to differences in factor endowments in different countries. The trade pattern is explained in terms of differences in relative factor endowments i.e., the relative resource supply in an economy. Ohlin's theory is also described as the factor endowment theory. Ohlin's theory is usually expounded in terms of a two-factor model with labour and capital as the two factors of endowments. Trade occurs due to the differences in factor endowments. Some countries have plenty of capital, while others have an abundance of labour. The Heckscher-Ohlin theorem postulates that countries rich in labour will export labour-intensive goods and countries with plenty of capital will export capital-intensive products.

Heckscher-Ohlin's theory explains the modern approach to international trade based on the following assumptions: -

- 1) There are two countries involved, countries A and B
- 2) There are two factors of production, labour and capital.
- 3) There are two goods, X and Y, of which X is labour-intensive, and Y is capital-intensive.
- 4) Country A is a labour-abundant country B is capital-rich.
- 5) There is perfect competition in both the commodity and factor markets.
- 6) All production functions are homogeneous of the first degree. Hence there are constant returns to the scale.
- 7) There are no transport costs or other impediments to trade.
- 8) Demand conditions are identical in both countries.

These assumptions have been made to explain the meaning of comparative price advantage or relative price difference and to deduce the major propositions of the factor endowment theory. Given these assumptions, Ohlin's thesis contends that, country exports goods which use relatively a greater proportion of its relatively abundant and thus cheap factors. It is implied that trade occurs because differences in relative commodity prices are caused by differences in relative factor prices (thus a comparative advantage) because of differences in the factor endowments among the countries.

The two countries, two commodities & two-factor model, imply that the capital-rich country will export capital-intensive commodities and the labour-rich country will export labour-intensive commodities. But the concept of a country being rich in one factor or the other is not very clear. Economists quite often define factor abundance in terms of factor prices. Ohlin himself has followed this approach. Alternatively, factor abundance can be defined in physical terms. In this case, physical amounts of capital & Labour are to be compared.

The Price Criterion of Relative Factor Abundance

According to the price criterion, a country with relatively cheap capital and labour relatively dear is regarded as relatively capital-abundant, irrespective of its ratio of total quantities of capital to labour compared with the other country. In symbolic terms, when:

$$(PK/PL)_E < (PK/PL)_I$$

Country E is relatively capital-abundant. (Here, P stands for factor price and K for capital, L for labour and E and I for the two respective countries.) Ohlin's theorem may be verified diagrammatically in Figure. 1.1

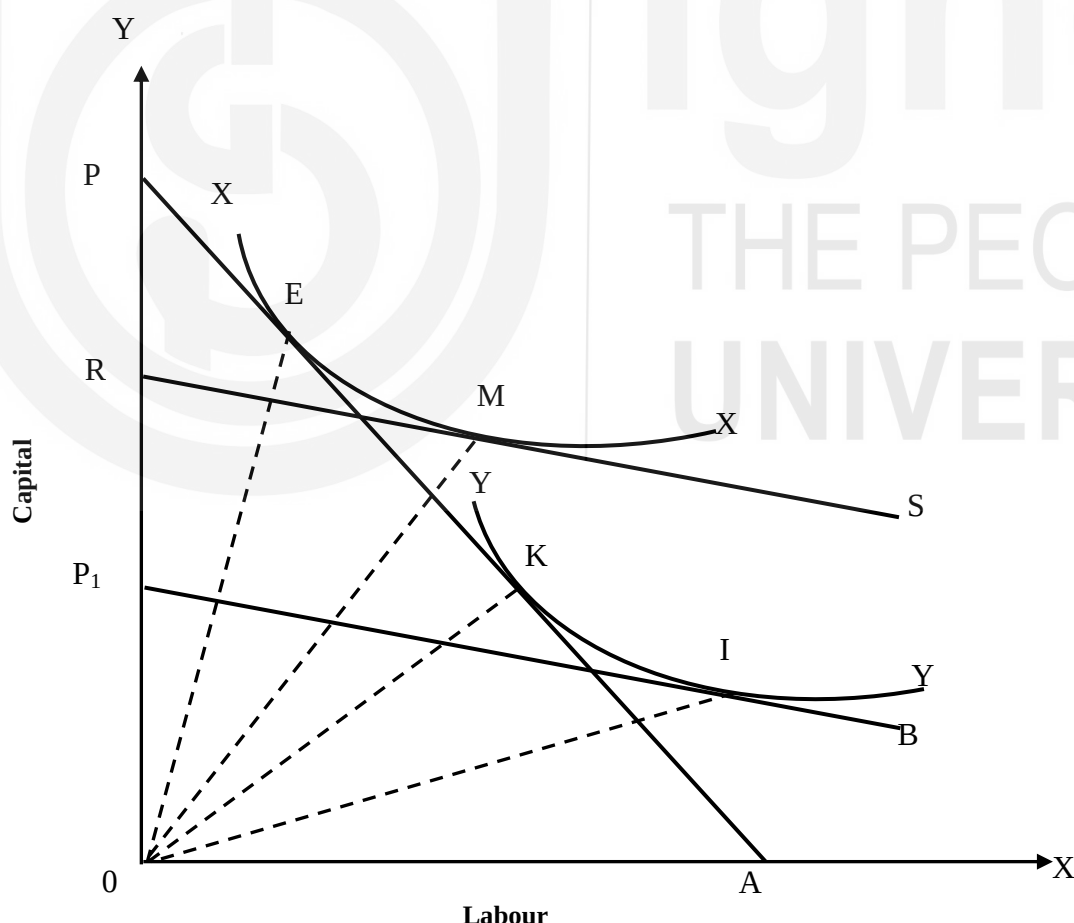


Figure 1.1 Diagrammatic explanation of H-O model

Let us take an example of the same two countries viz; England and India where England is a capital-rich country while India is a labour-abundant nation.

In the above diagram, XX is the isoquant (equal product curve) for the commodity X produced in England. YY is the isoquant representing commodity Y produced in India. It is very clear that XX is relatively capital intensive while YY is relatively labour intensive. The factor capital is represented on Y-axis while the factor labour is represented on the horizontal X-axis.

PA is the price line or budget line of the country England. The price line PA is tangent to XX at E. The price line PA is also tangent to YY isoquant at K. Point K will help us to find out how much capital and labour is required to produce one unit of Y in England.

P₁B is the price line of the country India, The price line P₁B is tangent to YY at I. The price line RS which is drawn parallel to P₁B is tangent to XX at M. This will help us to find out how much capital and labour is required to produce one unit of commodity X in India.

Under the given situations, the country England will choose the combination E. This means more specialisation in capital goods. It will not choose the combination K because it is more labour intensive and less capital intensive.

Thus, according to Ohlin, England will specialise in the production of goods X by using the cheap factor capital extensively while India will specialise on commodity Y by using the cheap factor labour available in the country.

Ohlin's theory concludes that the basis of international trade is the difference in commodity prices in the two countries. Differences in commodity prices are due to cost differences resulting from differences in factor endowments in two countries. A capital-rich country specialises in capital-intensive goods & exports them, while a labour-abundant country specialises in labour-intensive goods & exports them.

As the Ricardian framework of comparative advantage was further improved upon in the Heckscher-Ohlin model, similarly, the H-O model was further developed by Paul Samuelson. Hence the extension of the H-O model is also referred to as Heckscher-Ohlin-Samuelson (H-O-S) model. H-O model states that trade takes place due to the different relative prices of different goods in different countries. Heckscher-Ohlin-Samuelson model demonstrates how free movements of goods between countries may bring about the factor price equalization.

There are four main theorems in the H-O-S model.

- 1) The Heckscher-Ohlin theorem
- 2) The Stolper-Samuelson theorem
- 3) The Factor Price Equalization theorem
- 4) The Rybczynski theorem.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Point out five major assumptions of the Heckscher-Ohlin theory.

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2) What are the main conceptions of relative factor abundance?

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3) What are the extension theories of Heckscher-Ohlin theory?

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1.6 STOLPER – SAMUELSON THEOREM

Economists Paul Samuelson & Wolfgang Stolper have further contributed to this theory and have developed Stolper – Samuelson theorem. Stolper –Samuelson theorem explains the effect of change in relative product prices on factor allocation and income distribution. The Stopler-Samuelson theorem explains the relationship between changes in prices of goods and changes in prices of the factor of production such as wages and interest within the framework of the H-O theorem. The theorem states that if the price of the capital-intensive good increases then the price of capital will also increase, but the wages paid to labour will fall. Thus, if the price of machinery goods increases (assume machinery goods as capital-intensive) then the interest charged on capital would also increase. But at the same time, there will be a decrease in the wages of the labour. Similarly, if the price of labour-intensive goods increases then the wage rate would increase while the interest rate will fall.

The Stopler-Samuelson theorem was further extended by Jones, who developed the concept of 'magnification effect' for prices in the context of the H-O model. The magnification effect allows analysis of any change in the prices of both goods and provides information about the magnitude of the effects on wages and interest. Most importantly, the magnification effect allows one to analyse the

effects of price changes on real wages and real interest earned by workers and capital owners. This is crucial from the point of view of policy impact since real returns indicate the purchasing power of wages and rents after accounting for the price changes and thus are a better measure of well-being than simply the wage rate or rental rate alone.

This theorem is relevant in the age of globalization and trade liberalization. When trade liberalization occurs in a country, prices of goods change, and the magnification effect can be applied to seek an important result. A movement towards freer trade will cause the real return of the country's relatively abundant factor to rise, while the real return of the country's relatively scarce factor will fall. Thus, if the US and India are two countries that move towards free trade, and if the US is capital-abundant (while India is labour-abundant), then capital owners in the US will experience an increase in the purchasing power of their rental income while workers will experience a decline in the purchasing power of their wage income (i.e., they will lose). Similarly, workers will gain in India but capital owners will lose.

The reasons for this result are somewhat complex. When a country moves to free trade the price of its exported goods will rise while the price of its imported goods will fall. The higher prices in the export industry will inspire profit-seeking firms to expand production. At the same time, the import-competing industry suffering from falling prices, will want to reduce production to cut their losses. Thus, capital and labour will be laid-off in the import-competing sector but will be in demand in the expanding export sector.

However, a problem arises because the export sector is intensive in the country's (US) abundant factor, say capital – as per the H-O theorem. This means that the export industry wants relatively more capital per worker than the ratio of factors that the import-competing industry is laying off. In the transition, there will be an excess demand for capital, which will raise its price, and an excess supply of labour, which will bring down its price. Hence, the capital owners in both industries experience an increase in their rents while the workers in both industries experience a decline in their wages.

The theorem was originally developed to illuminate the issue of how tariffs would affect the incomes of workers and capitalists (i.e., the distribution of income) within a country because tariffs raise the domestic price of imported goods. However, the theorem is just as useful when applied to trade liberalization, as explained above. However, it should be kept in mind that these results have been derived in a model with only two goods and two factors that are perfectly mobile between sectors. This may not be valid in general. In particular, a factor employed in a sector where output declines (because of competition from imports) will suffer a loss in its real reward if it has no alternative source of employment.

1.7 FACTOR-PRICE EQUALIZATION THEOREM

According to the factor-price equalization theorem, when the prices of goods are equalized between countries due to international trade, the prices of the factors (i.e. capital and labour) also get equalized between countries. This implies that freer trade will equalize the wages of workers and the rentals earned on capital throughout the world in the ultimate analysis.

The theorem derives from the assumptions of the H-O model. The most critical is the assumption that the two countries share the same technology and that markets are perfectly competitive. In perfect competition, factors are paid based on the value of their marginal productivity, which in turn depends upon the prices of the goods. Thus, when prices differ between countries so will their marginal productivity, and hence so will their wages and rents. However, once prices of goods are equalized, as in free trade demonstrated earlier, the values of marginal products are also equalized between countries; hence, the countries must also share the same wage rates and rental rates.

However, it should be noted that the factor-price equalization is unlikely to apply perfectly in the real world. The H-O model assumes that technology is the same between countries to focus on the effects of different factor endowments. If production technologies differ across countries, as we assumed in the Ricardian model, then factor prices would not equalize once goods prices equalize.

Check Your Progress 4

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Point out the main argument of the Stopler-Samuelson theorem.

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- 2) Describe the magnification effect in the context of the H-O model.

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- 3) Explain the main statement of Factor Price Equalization theory?

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1.8 RYBCZYNSKI THEOREM

The Rybczynski theorem postulates that at constant commodity prices, an increase in the endowment of one factor will increase by a greater proportion of the output of the commodity intensive in that factor and will reduce the output of the other commodity. The theorem is useful in analyzing the effects of capital investment, immigration and emigration within the context of the H-O model.

Rybczynski's theorem can be understood diagrammatically with the help of figure 1.2. In Figure 1.2, EE reveals the labour constraints and while DD reflects the capital constraint. At point A, production takes place. Assume textiles as a labour-intensive good and software as a capital item and their production is depicted by C_1 C_2 for textiles and S_1 S_2 for software. Now, let us assume an increase in the wages of the labour, which causes an outward parallel shift in the labour constraint line from DD to D_1D_1 . Here, the production possibility frontier also shifts from point A to point B. Production of textiles, the labor-intensive good, will shift from C_1 to C_2 . Production of software, the capital-intensive good, will fall from S_2 to S_1 .

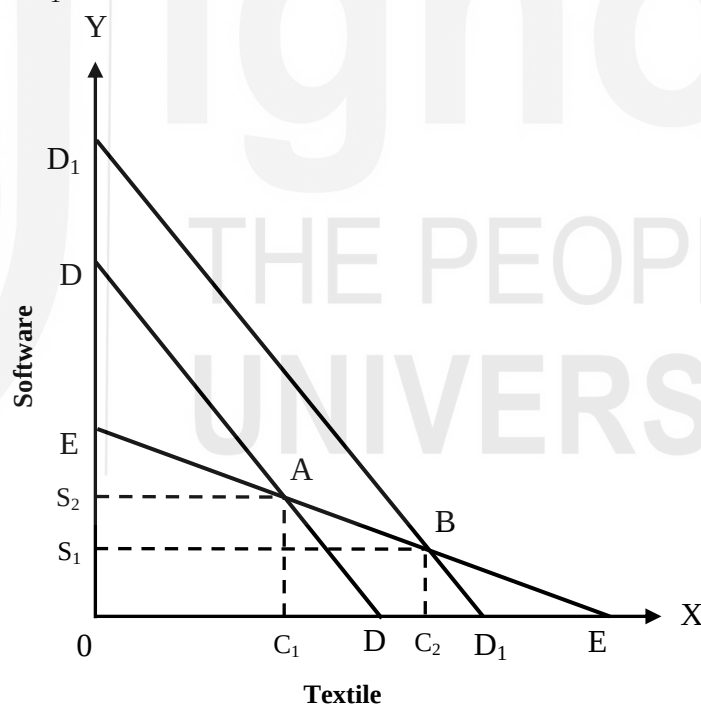


Figure 1.2 Diagrammatic representation of Rybczynski theorem

If the endowment of capital (interest rate) increases, the capital constraint will shift out, resulting in an increase in software production and a decrease in textiles production. As mentioned earlier, since the labour constraint curve is steeper than the capital constraint curve, the software is a capital-intensive item while textile is a labour-intensive item.

This means that in general, an increase in a country's endowment of a factor will cause an increase in the output of the good which uses that factor intensively and decrease the output of the other good.

Check Your Progress 5

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What does the Rybczynski theorem postulate?

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2) Point out the important factors which could be analysed in the context of Rybczynski's theorem.

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3) Explain the main statement of Factor Price Equalization theory?

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1.9 LET US SUM UP

After reading this unit, we have been exposed to the different international trade theories. We started with the theory of mercantilism, and then we understood the pure theory of international trade or absolute advantage theory coined by Adam Smith in 1776. After that, we learned the comparative advantage theory developed by David Ricardo in 1817. We learnt that both absolute advantage and comparative advantage theory considered labour as the only factor of production. Afterwards, we discussed Heckscher Ohlin's (H-O) Theory. Bertil Ohlin has drawn his ideas from Eli Heckscher's General Equilibrium Analysis. Hence, it is also known as Heckscher Ohlin (H-O) Theory. The unit also mentioned the extensions of the Heckscher Ohlin (H-O) model. Three more theories have been developed by extending the H-O model. The Stolper-Samuelson theorem explains the relationship between changes in prices of goods and changes in prices of a factor of production such as wages and interest within the framework of the H-O theorem. The theorem states that if the price of the capital-intensive good increases, then the price of capital will also increase, but the wages paid to labour will fall. Thus, if the price of machinery goods increases then the interest charged on capital would also increase. The unit also presents the Factor-Price Equalization Theorem. According to the factor-price equalization theorem when the prices of goods are equalized between countries due to international trade, the prices of the factors (i.e. capital and labour) also

get equalized between countries. This implies that freer trade will equalize the wages of workers and the rentals earned on capital throughout the world in the ultimate analysis. Finally, the unit concludes by discussing the Rybczynski theorem. The Rybczynski theorem postulates that at constant commodity prices, an increase in the endowment of one factor will increase by a greater proportion of the output of the commodity intensive in that factor and will reduce the output of the other commodity.

1.10 KEYWORDS

International Trade	The trade that takes place between buyers and sellers of two different nations is called international trade.
Mercantilism	It is the trade theory which postulates that countries should encourage exports and discourage imports. The theory argues that a nation should increase exports and reduce imports and export is the only way to accumulate wealth.
Comparative advantage	A country has a comparative advantage in producing a good 'A' if the cost of producing 'A' is lower at home than in the other country.
Factor Endowments	A country's endowments with resources like land, labour, capital etc.
Capital-Abundant Country	A country well-endowed with capital is referred to as capital abundant.
Magnification Effect	The magnification effect allows analysis of any change in the prices of both goods and provides information about the magnitude of the effects on wages and interest.

1.11 SOME USEFUL REFERENCES

- James Gerber. 2014 *International Economics*, 6th Ed., Pearson Education, USA.
- Krugman, Paul R. and Maurice Obstfeld, 1997. *International Economics: Theory and Policy*, 4th Ed., Addison-Wesley, Massachusetts, USA.
- Sodersten, B. 1980. *International Economics*, Macmillan London.
- Sodersten, B. and Reed, G. 1994. *International Economics*, 3rd Ed., Macmillan London.
- Salvatore, D. John 2004. *International Economics*, 8th Ed., Wiley and Sons, Inc. New Jersey.

1.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Classical and Neo-
Classical Theories of
International Trade

Check Your Progress 1

- 1) Refer to Section 1.1
- 2) Refer to Section 1.2

Check Your Progress 2

- 1) Refer to Sections 1.3 and 1.4
- 2) Refer to Sections 1.3 and 1.4
- 3) Refer to Section 1.4

Check Your Progress 3

- 1) Refer to Section 1. 5
- 2) Refer to Section 1.5
- 3) Refer to Section 1.5

Check Your Progress 4

- 1) Refer to Section 1. 6
- 2) Refer to Section 1.6
- 3) Refer to Section 1.7

Check Your Progress 5

- 1) Refer to Section 1.8
- 2) Refer to Section 1.8

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UNIT 2 GAINS FROM TRADE *

Structure

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Meaning of Gains from Trade
- 2.3 Sources of Gains
- 2.4 Factors Determining Size of Gains
- 2.5 Production Possibilities Curve in International trade
- 2.6 Measurement of Gains from Trade
- 2.7 Potential and Actual Gain
- 2.8. Free Trade versus No Trade
- 2.9 Static and Dynamic Gains
- 2.10 Let us Sum up
- 2.11 Key Words
- 2.12 Some Useful books
- 2.13 Answers/Hints to Check Your Progress Exercises

2.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain the meaning of gains from international trade and sources of such gains;
- identify the conditions under which two countries can gain from trading with each other;
- elucidate and illustrate how the terms of trade determine the extent to which each country specializes; and
- explain and illustrate the mutual benefits of trade.

2.1 INTRODUCTION

As we have discussed in the previous unit, international trade refers to the voluntary exchange of goods and services between different countries. Trade between countries will only occur if they consider it beneficial to their interests.

* Contributed by Dr. Shyam Sundar, Sr. General Manager - Policy Advisory, Mahindra Group

The previous unit briefly discussed about why do countries trade? In this unit, we shall elaborate the concept of gains from international trade.

In this connection, the meaning of gains from trade will be discussed followed by sources of gains and factors determining gains from trade. Thereafter, we shall discuss the concept of the Production Possibilities Curve which shows all the various alternative combinations of the two commodities that a country can produce by completely utilising all its factors of production with the best available technology.

Subsequently, we will explain how gains from trade can be measured. Which imply how each nation gains from international trade by exchange and specialisation in the production?

Lastly, we analyse the various situations of the potential and actual gain, Free Trade vs. no trade and static and dynamic gains.

2.2 MEANING OF GAINS FROM TRADE

Gains from international trade refer to the various benefits which country derived by way of trading of goods and services with other countries. Such benefits from trade are because of international division of labour and specialisation. Countries trade with each other because trade is beneficial to all. Basic motivation of trade is the gain or benefit that nations accrue over the period. Each trading country gains when the total output increases as a result of the division of labour and specialisation. In the case of autarky or isolation, the benefits of the international division of labour do not flow between nations.

Malthus held that 'the gain from trade consisted of the increased value which results from exchanging what is wanted less for what is wanted more, and that 'international trade, by giving us commodities much better suited to our wants and tastes than those which had been sent away, has decidedly increased the exchangeable value of our possessions, our means of enjoyment, and our wealth'.

In simple words, gains from trade refers to additional production and consumption effects that countries can achieve by way of entering trade with each other. To sum-up, gains from international trade are of two types–i) gain from exchange and ii) gain from specialisation in production. Approaches preferred by various economists will be discussed in subsequent sections.

2.3 SOURCES OF GAINS

As per classical theory, major source of gains from international trade arises out of specialisation based on the principle of comparative cost advantage. While pointing out the 'division of labour', Adam Smith has held that it is limited by the size of the market. Upon expansion of the market size because of international trade, the scope for large-scale production also increases which in turn increases the scope for complex division of labour and specialisation. In short, international

trade leads to enlargement of the market which further leads to specialisation and division of labour. This process further results in an increase in output per unit of input.

The comparative cost theory exhibits increased world production as gains from international trade in the real-world scenario. Each trading country gains by way of getting relatively more, better and cheaper goods. In the entire process no country lose by having less than what they needed.

In this way we can say that gains from trade are results of specialization in production from the division of labour, economies of scale, and agglomeration along with relative availability of factor resources.

When there is free trade, goods and services produced all over the world are available to people everywhere. In other words, international trade makes available to the people of a country, a galaxy of goods and services at the most competitive prices. A country may not have the factor endowments or technological capability to produce certain goods. If there is no trade with other countries, it will have to do without such goods but through international trade, it can procure them.

The gains from international trade may be summed up as follows:

- i) Expansion of the size of the market
- ii) Division of labour
- iii) Gains from specialisation
- iv) Gains from increased product variety
- v) Gains from competition
- vi) Gains from increased economies of scale
- vii) Productivity gains.

2.4 FACTORS DETERMINING SIZE OF GAINS

Differences in cost ratios: if country A has a comparative advantage in the production of wheat and country B has a comparative advantage in the production of cotton, both countries will gain from trade. The size of the gain will depend on the cost of production of each commodity in both countries. The gain from international trade depends upon the cost ratios of differences in comparative cost ratios in the two trading countries. The larger the difference between exchange rate and cost of production the larger the gains from trade and vice versa.

- 1) **Reciprocal demand:** The reciprocal demand means the relative strength and elasticity of demand of one country for the product of the other in exchange for its product.

- 2) **Level of income:** The level of money income of a country is another factor which determines the gains and the share of trade. A country whose goods have constant demand in other countries will have a high level of money income.
- 3) **Terms of trade (also called “trading price”):** It is the most important factor which determines the gains from trade. The international terms of trade refer to the rate at which one commodity of a country is exchanged for another commodity of the other country. If the cost ratio and terms of trade are closer to each other, more will be the gains from the trade of the participating countries.
- 4) **Productive efficiency:** An increase in the productive efficiency of a country also determines its gains from trade. It lowers the costs of production and prices of goods in the home country. As a result, the other country gains by importing cheap goods and its terms of trade improve.
- 5) **Nature of commodities exported:** Another factor is the nature of commodities exported by a country. A country that exports mainly primary products has unfavourable terms of trade. Consequently, its gains from trade will be smaller. On the contrary, a country exporting manufactured goods has favourable terms of trade and its gain from trade will be larger.
- 6) **Technological conditions:** In a country which is technologically advanced and has an abundance of capital, its volume of foreign trade will be large and so it will gain from international trade. On the other hand, if a country is technologically backward with abundant labour, its volume of foreign trade will be small and so will be gains from trade.
- 7) **Size of the country:** If a country is small, it is relatively easy for them to specialize in the production of one commodity and export the surplus production to a large country and can get more gains from international trade. Whereas if a country is large, then they must specialise in more than one good because the excess production of only one commodity cannot be exported fully to a small-sized country as the demand for the commodity will reduce very frequently. So, the smaller the size of the country, the larger the gains from trade.
- 8) **Factor availability:** International trade is based on specialization and a country specializes depending upon the availability of factors of production. It will increase the domestic cost ratios and thereby the gains from trade.
- 9) **Productive Efficiency:** An increase in the productive efficiency of a country also determines its gains from trade. A highly efficient country can keep the cost of production and price of the goods lower than other countries.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Why do countries trade with each other? What are the gains from trade?

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2) Examine the main source of gains from trade?

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3) How do the gains from trade help an economy grow?

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2.5 PRODUCTION POSSIBILITIES CURVE IN INTERNATIONAL TRADE

Production Possibilities Curve (also known as production frontier, transformation curve, product substitution curve or an opportunity cost curve) shows all the various alternative combinations of the two commodities that a nation can produce by fully utilizing all its factors of production with the best available technology.

The slope of the production possibilities curve refers to the marginal rate of transformation (MRT) or the amount of a commodity that the nation must give up to get one additional unit of the second commodity.

If the nation faces constant costs or MRT, then its production possibilities curve is a straight line with an (absolute) slope equal to the constant opportunity costs or MRT and the relative commodity price in the nation.

The country lacks the capacity beyond the limit specified by the production possibility curve. It shows the production frontier of the country.

Producing output beyond the production frontier is not possible. If the output of the two or one of the two commodities is below the production frontier, that indicates unemployment or excess capacity. To be more specific, consider a situation when the production of Y commodity is reduced to produce more units of X commodity, it signifies that Y has been transformed or substituted into X-commodity. Therefore, it is called an opportunity cost curve, transformation curve or product substitution curve.

Example: When all the available resources are employed in production of only one commodity say commodity X, commodity X can be produced at maximum possible level without any production of another commodity say Y. On the opposite, if all the resources are utilised in the production of Y, the country will be able to produce maximum quantity of Y commodity with no output of X commodity. Between these two extreme situations, there can be various production possibilities involving more or fewer quantities of the two commodities. If the production of X is to be increased, there will be a diversion of resources from the production of Y to the production of X, resulting in reduced production of Y.

MRT is the ratio of a change in the quantity of commodity Y to a change in the quantity of X commodity.

$$MRT_{xy} = \delta y / \delta x$$

Since additional production of X involves reduced output of Y, the MRT is negative. It signifies that the production possibility curve or opportunity cost curve slopes negatively, or it slopes downwards from left to right.

The MRT_{xy} can be expressed also as a ratio of the marginal cost of X to the marginal cost of Y.

It can be determined as below:

$$\delta C = MC_x \cdot \delta x + MC_y \cdot \delta y$$

Where δC = Change in cost, δx = Change in the quantity of X commodity, δy = Change in the quantity of Y commodity. MC_x and MC_y are the marginal costs of X and Y commodities respectively.

If

$$\delta C = 0$$

$$MC_x \cdot \delta x + MC_y \cdot \delta y = 0$$

$$MC_x \cdot \delta x = - MC_y \cdot \delta y$$

$$\frac{MC_x}{MC_y} = - \frac{\delta y}{\delta x}$$

$$\text{or } MRT_{xy} = - \frac{\delta y}{\delta x} = \frac{MC_x}{MC_y}$$

If the production is governed by constant returns, the MC_x relative to MC_y remains unchanged or MRT_{xy} remains the same. It means the slope of the production possibility curve or opportunity cost curve is the same and it is a negatively sloping straight line. If the production is governed by diminishing returns, MC_x rises relative to the MC_y . It signifies that the slope or MRT_{xy} increases. In such a situation, the opportunity cost curve is a negatively sloping concave curve to the origin. If the production is governed by increasing returns, the MC_x decreases relative to the MC_y . The slope or MRT_{xy} decreases. In this

case, the opportunity cost curve is a negatively sloping convex curve to the origin. These cases are depicted in Fig. 2.1 (a), 2.1 (b) and 2.1(c) respectively.

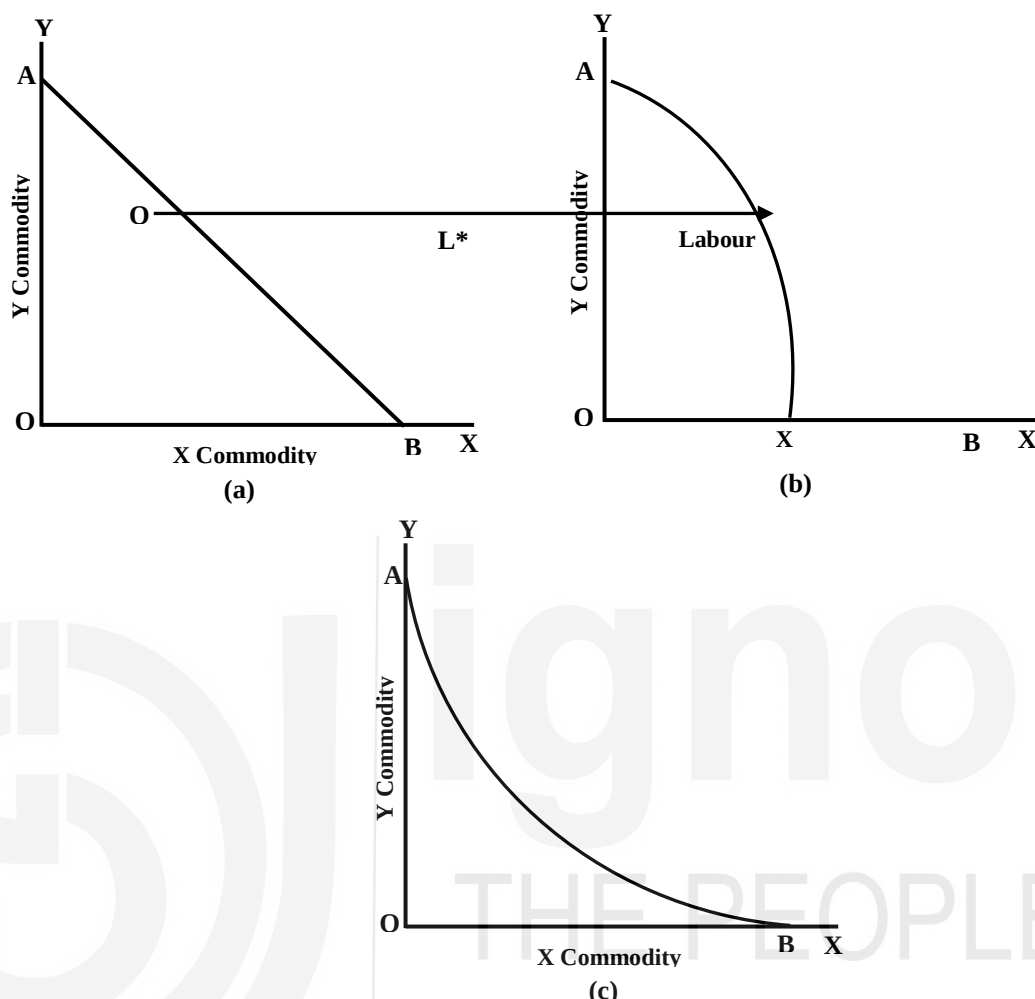


Figure 2.1(a), 2.1(b) and 2.1 (c)

The opportunity cost curve simply indicates the alternative production possibilities. It does not show what combinations of the two commodities will be produced.

In Fig. 2.1, AB is the production possibility curve or the opportunity cost curve. In Fig. 2.1 (a), the opportunity cost curve AB is the negatively sloping straight line. Along this curve, $MRT_{xy} = -\delta y / \delta x = MC_x / MC_y$ remains unchanged due to constant opportunity cost conditions (constant return)

In Fig. 2.1 (b), the opportunity cost curve AB is negatively concave. Along it, $MRT_{xy} = -\delta y / \delta x = MC_x / MC_y$ increasing opportunity cost condition (diminishing returns).

In Fig. 2.1 (c), the opportunity cost curve AB is a negatively sloping convex curve to the origin on account of decreasing opportunity cost condition (increasing returns).

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Why is the production possibilities curve of each nation usually different?

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2) Is trade still possible between two nations if they have identical production possibilities curves?

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3) Under what conditions can no trade take place between two nations with different production possibility curves?

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4) What happens if two nations are identical in every respect?

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2.6 MEASUREMENT OF GAINS FROM TRADE

Jacob Viner pointed out that the gains from trade were measured by the classical economists in terms of:

- i) An increase in national income,
- ii) Differences in comparative costs, and
- iii) Terms of trade.

Some approaches to the concept of gains from trade and their measurement are discussed below:

1) **Adam Smith's Approach:**

According to Adam Smith, international trade helps to improve the productive capacity of trading nations. It increases value of the product as excess of a product in domestic market can be exported. A nation can also import products which are high in demand in domestic market. Thus, gains

from international trade are in form of maximum welfare through maximum possible export earnings.

A country reaches the point of optimum allocation as each country specializes in production of a commodity in which it has cost advantage. Specialization results in economies of scale i.e., reduced cost of the product.

2) Ricardo-Malthus Approach:

In unit 1, we have discussed about Comparative advantage theory. You can recall that Ricardo asserted that principle of comparative costs in international trade results in saving costs and resources. In the words of David Ricardo, "The advantage to both places is not that they have any increase in value but with the same amount of value they are both able to consume and enjoy an increased quantity of commodities."

Malthus had expressed in this regard views similar to those of Adam Smith. The gain from trade, according to him, consists of "the increased value, which results from exchanging what is wanted less for what is wanted more." The international exchange on this basis increases the "exchangeable value of our possession, our means of enjoyment and our wealth."

The Ricardo-Malthus approach to gains from trade was illustrated by Ronald Findlay in Fig. 2.2.

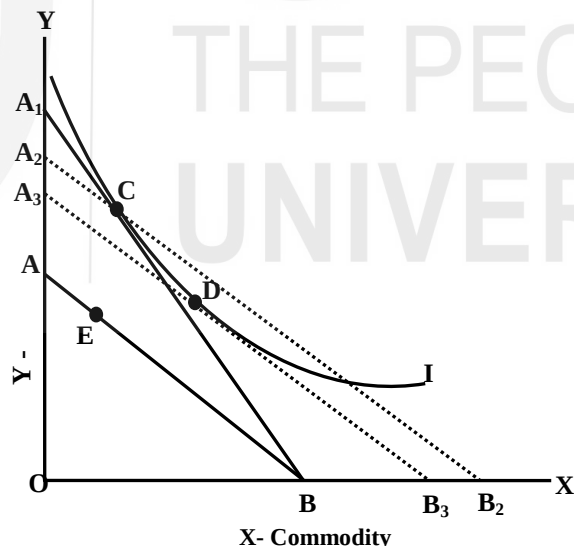


Figure 2.2: The Ricardo-Malthus approach to gains from trade

In Fig. 2.2., AB is the production possibility curve under constant cost conditions before the trade. Suppose the production takes place at E. If this country enters into trade, the international exchange ratio line shifts to A₁B and production of two commodities X and Y now takes place at C. The production at point C will be possible if the labour input increases an extent that the production possibility curve shifts to A₂B₂. The gain from trade will be measured by BB₂/OB.

Malthus criticized this measure of gain from trade as exaggerated. According to him, if the production possibility curve shifts to A_2B_2 , point C cannot be the point of equilibrium. The relative prices along A_2B_2 are more favourable to the export good X than along the line A_1B . Some points to the right of C rather than C itself would be preferable to the community. Hence the gain from trade along line A_1B cannot be measured by an increase in the input of labour in the ratio BB_2/OB . It tends only to overstate the gain from trade.

Ronald Findlay attempted a modification over the Ricardian measure of gain from trade by introducing in this analysis the community indifference curve. Given the community indifference curve I, the equilibrium does not take place at the Ricardian trade equilibrium position C but at D where the production possibility curve A_3B_3 became tangent to the community indifference curve I.

At point D, everyone in the community is better off than at C. The gain from trade in this situation is BB_3/OB rather than BB_2/OB . The measure of gain from trade BB_3/OB vindicates the Malthusian criticism that the Ricardian measure of gain from trade was an over-estimation.

3) J.S. Mill's Approach:

Ricardian approach did not explain the distribution of gains from trade among the trading countries. J.S. Mill in his approach attempted to analyse both the gains from trade and the distribution thereof among the trading countries. He stressed on the concept of reciprocal demand.

According to him, reciprocal demand determines terms of trade, which is a ratio of quantity imported to the quantity exported by a given country. The terms of trade determine the distribution of gains of trade between the trading partners.

Suppose in country A, 1 unit of labour can produce 10 units of X and 10 units of Y so that the domestic exchange ratio in country A is: 1 unit of X = 1 unit of Y. In country B, 2 units of labour can produce 12 units of X and 18 units of Y so the domestic exchange ratio in this country is: 1 unit of X = 1.5 units of Y. The domestic exchange ratios set the limits within which the actual exchange ratio or terms of trade will get determined.

The reciprocal demand or the strength of the elasticity of demand of the two trading countries for the products of each other will decide the actual rate of exchange of two commodities. If A's demand for commodity Y is less elastic, the terms of trade will be closer to its domestic exchange ratio: 1 unit of X = 1 unit of Y. In this case, the terms of trade will be favourable for country B and against country A.

The gain will be more for B than for A. On the contrary, if B's demand for X commodity is less elastic, the terms of trade will be closer to the domestic exchange ratio of country B: 1 unit of X = 1.5 unit of Y. The terms of trade, in this situation, will be favourable for A and against B. Country A will have a larger share out of the gains from trade than country B.

The distribution of gains from trade can be explained in terms of the Marshall-Edgeworth offer curve through Fig. 2.3.

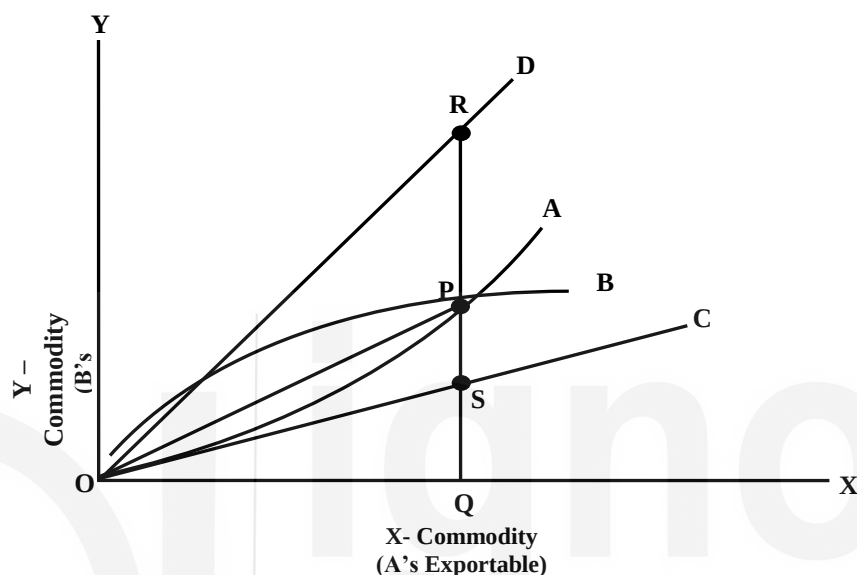


Figure 2.3: Marshall-Edgeworth offer curve

In Fig. 2.3., OC and OD are the domestic exchange ratio lines of countries A and B respectively. OA is the offer curve of country A and OB is the offer curve of country B. The exchange takes place at P where the two offer curves cut each other. Country A imports PQ quantity of Y and exports OQ quantity of X.

The terms of trade for country A at P = $(Q_Y/Q_X) = (PQ/OQ) = \text{Slope of Line OP}$. If the line OP gets closer to OD, the terms of trade become favourable to country A and unfavourable to country B. On the opposite, if the line OP gets closer to the line OC, the domestic exchange ratio line of country A, the terms of trade turn against country A and become favourable to country B.

Country A was willing to exchange before trade SQ units of Y for OQ units of X. After the trade, it gets PQ units of Y for OQ units of X. Therefore, the gain from trade for country A, out of the total trade gain of RS, amounts to $PQ - SQ = PS$ units of Y. In the case of country B, RQ units of Y were being exchanged for OQ units of X before the trade.

However, after trade it has to part with only PQ units of Y to import OQ units of X. Therefore, the gain from trade for this country amounts to $RQ - PQ = RP$ units of Y. As the point of exchange, P gets closer to the line OD, the share of country A in the gain from trade will rise and that of country B will fall and vice-versa.

4) Taussig's Approach:

According to Taussig, trading country gains from trade in form of a rise in income. As the demand for labour increases because of increase in exports, employers offer higher wages to attract more labour in this industry. The spill over effect of rise in wages in one industry is that other industries will also have to offer higher wages to retain talent. Thus international trade, according to Taussig will lead to a general rise in money incomes. A higher level of income due to trade enables the people of a country to make larger purchases of both domestically produced and imported goods and reach a higher level of welfare.

5) Modern Approach:

The modern theorists considered the gains from trade as the gains resulting from:

- i) Gains from exchange and
- ii) Gains from specialisation.

These two gains together constitute the gains from international trade. When trade takes place between countries, consumers enjoy a higher level of satisfaction, partly because of improvement in terms of trade and partly on account of greater specialisation in the use of economic resources of the country. These two types of gains from trade can be shown in Fig. 2.4.

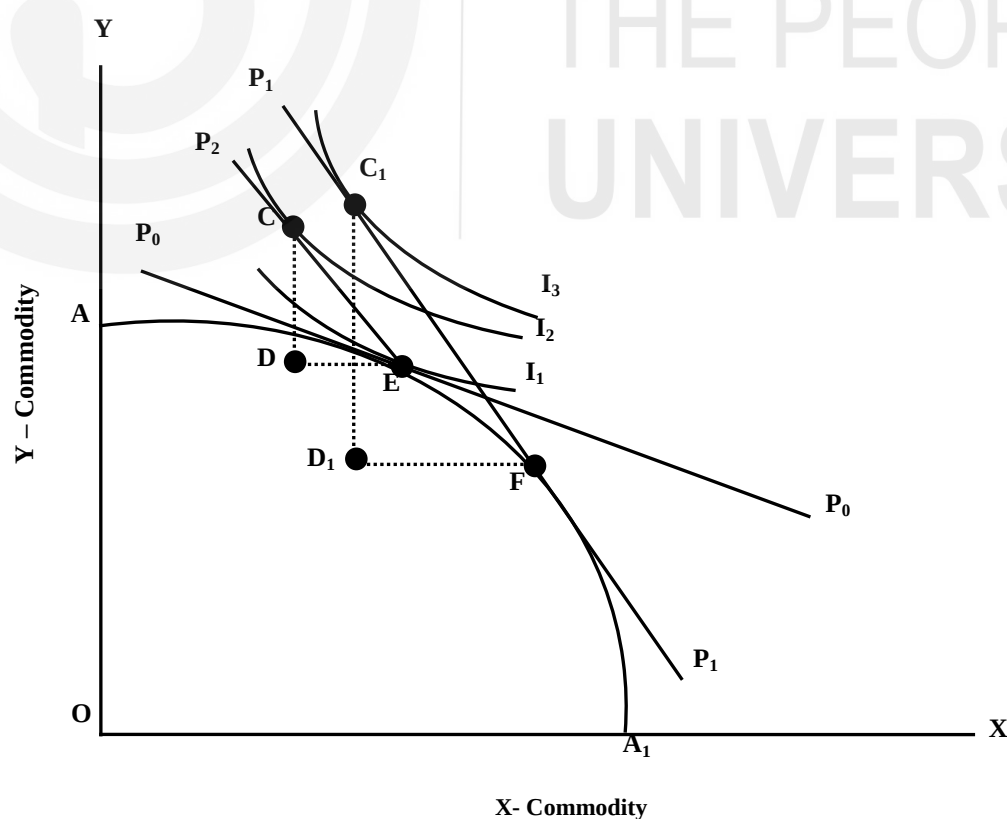


Figure 2.4: Gains from trade

In Fig. 2.4., X-commodity is measured along the horizontal scale and Y-commodity is measured along the vertical scale. AA_1 is the production possibility curve. P_0P_0 is the domestic price ratio line. It is tangent to the production possibility curve at E. Thus, E is the point of production equilibrium in the absence of trade.

E is also the point of consumption equilibrium because P_0P_0 is tangent to the community indifference curve I_1 at this point. When trade commences, P_1P_1 is the international exchange ratio line, which is tangent to the production possibility curve at F and the community indifference curve I_3 at C_1 .

Thus, F is the point of production and C_1 is the point of consumption. After trade takes place, D_1F of X-commodity is exported and C_1D_1 quantity of Y-commodity is imported. The trade causes two types of shifts in the country. First, the production point shifts from E to F. It occurs because of specialisation in the production of X-commodity and specialisation in factor use. This can be called the production effect. Second, the point of consumption shifts from E at I_1 to C_1 at the higher community indifference curve I_3 . It means an increase in the satisfaction of the commodity. This can be called the consumption effect. Both together signify the gain from international trade.

If a line P_2E is drawn parallel to P_1P_1 from the original equilibrium situation E, it signifies that there is no change in production, but the consumption equilibrium shifts from E to C at a higher community indifference curve I_2 . In this situation, the CD quantity of Y is imported at a lower international price of Y. The quantity of X-commodity exported in exchange for CD quantity of Y is DE. Although production is the same as at point E, the consumption equilibrium shifting from E to C signifies the gain from trade. This is the trade gain from the exchange.

After the trade, as the specialisation in production and optimum factor use takes place, the production equilibrium shifts from E to F along the same production possibility curve and the consumption equilibrium shifts to C_1 . In this situation, C_1D_1 quantity of Y is imported and D_1F quantity of X is exported. As a result of specialisation in production after the trade, the shift in consumption equilibrium from C to C_1 reflects the trade gain from specialisation.

To sum up, the total gain from trade is comprised of gain from the exchange and the gain from specialization. The total gain from trade can be measured by the movement from E to C_1 . This movement takes place in two steps—the movement from E to C is the gain from the exchange and the movement from C to C_1 is the gain from specialization.

Another Example:

Gains from Trade

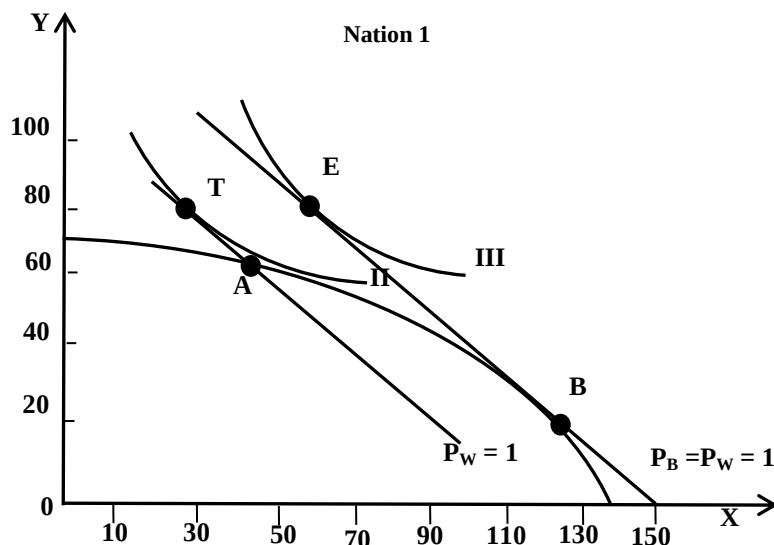


Figure 2.5: Gains from specialisation in production

If Nation 1 could not specialize in the production of X with the opening of trade but continued to produce at point A, Nation 1 could export 20X in exchange for 20Y at the prevailing world price of $P_W = 1$ and end up consuming at point T on indifference curve II. The increase in consumption from point A (in autarky) to point T represents the gains from exchange alone. If Nation 1 subsequently did specialize in the production of X and produced at point B, it would then consume at point E on indifference curve III. The increase in consumption from T to E would represent the gains from specialization in production.

6) Gains from Trade for Large Country versus Small Country:

H.R. Heller explained how entire gains go to a small country instead of large country under the conditions of constant opportunity cost and unchanged terms of trade. This case can be explained in Fig. 2.6.

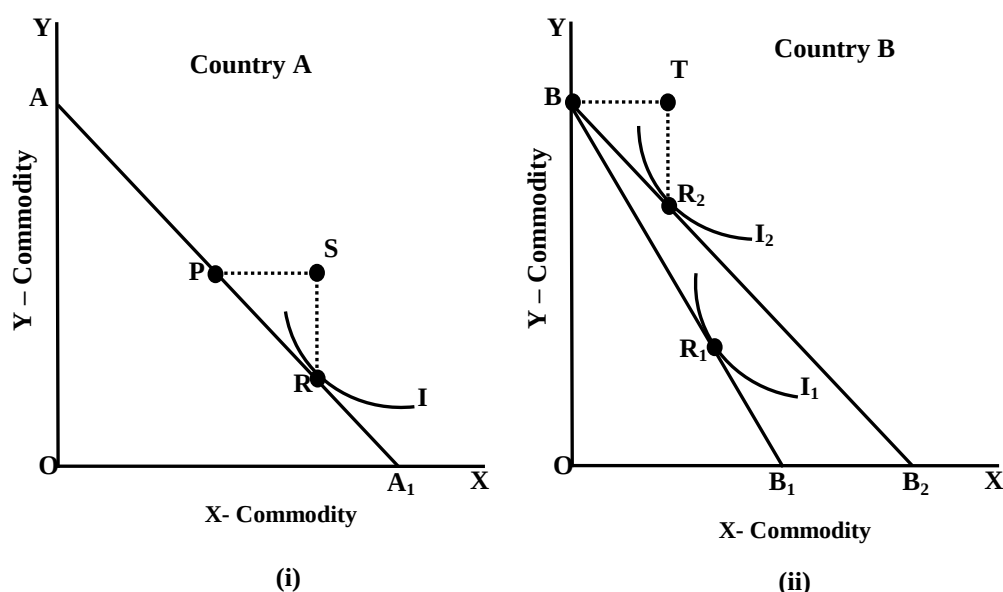


Fig. 2.6. (i) & 2.6. (ii): Gains from Trade for Large Country and Small Country

In Fig. 2.6. (i) For a large country A, the production possibility curve under the conditions of constant costs is AA_1 . In the absence of trade, consumption and production take place at R where the community indifference curve I is tangent to the production possibility curve. After trade takes place, there is no change in terms of trade for country A so that the international price ratio line remains AA_1 . This country will, however, modify its production pattern in such a way that some imports are made from country B. It may decide to move to P where it exports PS quantity of X commodity and imports SR quantity of Y. Since the terms of trade remain unchanged for country A, it fails to make any gain from trade.

In Fig. 2.6. (ii), for the small country B, the production possibility curve or domestic price ratio line under constant cost conditions is BB_1 . Its tangency with the community indifference curve I_1 shows that production and consumption equilibrium in this country, in the absence of trade, takes place at R_1 . As trade commences, this country specialises completely in the production of Y commodity. The international price ratio line is BB_2 , which is parallel to AA_1 . This country produces at B. The consumption equilibrium occurs at R_1 .

So, after trade it exports $TR_2 (= SR)$ of Y commodity to country A and imports $BT (= PS)$ quantity of X from country A. The movement from R_1 to R_2 in country B reflects the gain from specialisation and exchange to the small country B from the international trade. Since this country can import X-commodity at a lower international price, the terms of trade turn in favour of it. That also shows that the gains from trade go to small country B alone and the large country goes without any gain from trade.

2.7 POTENTIAL AND ACTUAL GAIN FROM TRADE

Domestic cost ratios of producing two commodities X and Y determines the potential gain from trade for the two trading countries A and B.

Thus,

$$G_p = \left(\frac{C_X}{C_Y} \right)_A - \left(\frac{C_X}{C_Y} \right)_B$$

Where G_p = potential gain C_X = cost per unit (money or labour) for X commodity, C_Y = Cost per unit (money or labour) for Y commodity. The subscripts 'A' and 'B' signify the two countries.

The actual gain from trade G_a , on the other hand, is determined by the difference in price ratio of the two commodities X and Y in the two countries.

Thus,

$$G_a = \left(\frac{P_X}{P_Y} \right)_A - \left(\frac{P_X}{P_Y} \right)_B$$

Under the conditions of perfect competition and free trade, the cost ratio equals the price ratio of two commodities in each country such that

$$\frac{C_X}{C_Y} = \frac{P_X}{P_Y}$$

Consequently the actual gain becomes equal to potential gain

$$G_p = G_a$$

In absence of tariffs and other trade restrictions, actual gains are equal to potential gains of the country. In case of presence of imperfect competition or trade and tariff restrictions, the price ratio (P_X/P_Y) is more than the cost ratio (C_X/C_Y) of a country. Thus, the actual gain from trade exceeds the potential trade gain ($G_a > G_p$).

2.8 FREE TRADE VERSUS NO TRADE

When there are no tariffs or trade restrictions, it is called a situation of free trade. In situation of free trade, domestic factor and product prices get equalised with international prices.

Assumptions:

The proposition that free trade is superior to no trade is proved based on the following assumptions:

- a) There is a situation of perfect competition in the market.
- b) The government does not interfere in trade i.e. there are no tariffs, quotas and subsidies.
- c) There are no transportation costs.
- d) The given country is small in size and has no monopoly power in trade.
- e) The factors of production are fixed in supply.
- f) The technology is such that the production possibility curve is concave to the origin.

Based on the assumptions given above, it is possible to show that free international trade is much superior to autarchy (absence of trade). The diagram, to demonstrate it, is adapted from the diagram given by Jagdish Bhagwati.

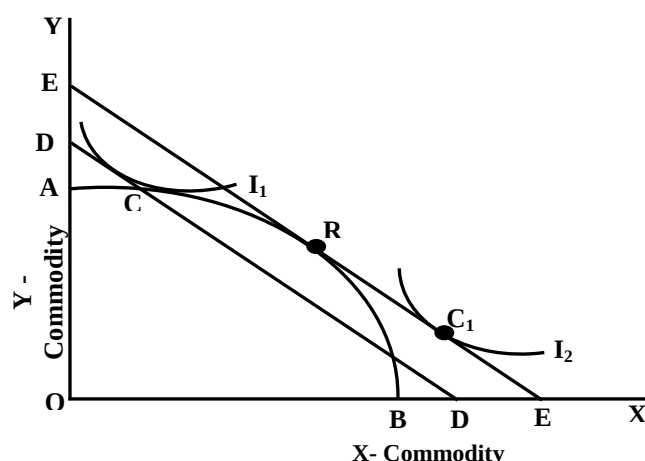


Figure 2.7: Benefits of free trade vs. no trade

In Fig. 2.7 AB is the production possibility curve of the home country. In the absence of trade, the domestic price ratio is given by the line DD. C is the point of production and consumption equilibrium. As the trade commences and there is no restriction on trade, the international price ratio is given by the slope of the line EE which runs parallel to DD. The line EE represents the Consumption Possibility Curve.

EE determines the new availability frontier in the country. Point R, where the consumption possibility curve is tangent to the production possibility curve, represents the most efficient production point. Consumption point on the other hand is determined at C_1 where the international price ratio line EE is tangent to the higher community indifference curve I_2 . Since after free trade, the production is optimised at R and consumption is optimised at C_1 , it follows that free trade is superior to no trade.

2.9 STATIC AND DYNAMIC GAINS

The gains from international trade are of two types:

1) Static Gains & Dynamics Gains

The static gains from trade are as under:

- i) **Division of labour, specialization and Expansion in Production:**
International trade accord the benefit of division of labour and specialization. It ensures that all the available productive resources in the country are optimally utilized leading to expansion in production of the firm, country and the world.
- ii) **Improved Welfare:** Increased production because of international trade leads to improved welfare in the trading countries. Quality of the products also improves due to specialisation by a country. Quality along with easy access at a lower costs directly boosts welfare of the people.

As Ricardo points out "The extension of international trade very powerfully contributes to increasing the mass commodities and, therefore, the sum of enjoyments."

- iii) **Increase in National Income:** International trade leading to expansion of production in the trading countries helps to generate more and more employment opportunities. Increase in production and employment leads to a rise in the national income of the trading countries.
- iv) **Vent for Surplus:** According to Adam Smith, international trade leads to the fullest utilisation of productive resources of the country. The surplus created because of increased efficiency and increased production can be utilized through sales in the foreign market.

2) Dynamic Gains from Trade:

The major dynamic gains from international trade are as follows:

- i) **Technological Development:** International trade facilitates transfer of advanced technology from developed to less developed countries. It stimulates innovations and inventions which in turn bring efficiency in production and reduces overall cost. Latest and advanced techniques are transferred to other countries through trade.
- ii) **Increased Competition:** Along with other benefits, international trade brings in efficiencies in all the industries in trading countries. It creates the environment of competition ultimately leading to increased quality of goods.
- iii) **Expansion of Market:** International trade leads to expansion in the size of the market. It encourages producers to expand the scale of production which in turn leads to the increased volume of investment and employment. Increased investment and employment ultimately lead to expansion in the market.
- iii) **Increase in Investment:** As the demand for home-produced goods increases due to international trade, there is a strong impetus for investment. The growth of the export sector leads to the expansion of several allied ancillary industries creating more and more opportunities for investment. There is also a substantial increase in foreign direct investments in the export sector of the economy.
- v) **Optimum Use of Resources:** International trade leads to more efficient use of all the productive resources.

Ellsworth and Clark Leith summed up the dynamic gain from trade in these words, "Trade is a dynamic force that stimulates innovation. New ways of producing and organising production are spread to the local economy through

trade and the competitive force of trade stimulates the adoption of cost-saving techniques. Trade also makes possible economical local production of many goods that would be prohibitive to produce locally."

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) In what circumstances might a country NOT benefit from trade with another country?

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- 2) What are static and dynamic gains from trade?

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- 3) Explain Mill's approach to measuring gains from trade.

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- 4) Why is 'free trade' between two countries is considered to be better than 'no trade'?

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2.10 LET US SUM UP

After reading this unit, we learnt :

- a) To maximize the value of its output, a country must be producing a combination of goods and services that lies on its production possibilities curve.
- b) Suppose two countries each produce two goods and their opportunity costs differ. If this is the case, there is an opportunity for trade between the two countries that will leave both better off.
- c) International trade leads countries to specialize in goods and services in which they have a comparative advantage.

- d) The terms of trade determine the extent to which each country will specialize. Each will increase the production of the good or service in which it has a comparative advantage up to the point where the opportunity cost of producing it equals the terms of trade.
- e) Free international trade can increase the availability of various goods and services in the countries that participate in it. Trade between the countries allow consuming combinations of goods and services which they would not be able to produce.
- f) While free trade increases the availability of total quantity of goods and services to each country, in short run, there might be winners and losers.

2.11 KEYWORDS

Autarky	The absence of trade; economic isolation
Absolute Advantage	The greater efficiency that one nation may have over another in the production of a commodity. It describes a situation in which a country can produce more of a good or service than any other with the same quantity of resources. According to Adam Smith, this is the basis for trade.
The basis for Trade	The forces (absolute advantage for Adam Smith and Comparative Advantage for Ricardo).
Specialisation	When a country allocates most or all its resources towards the production of a particular good or service. For example, Bangladesh specializes in producing textiles.
Gains from Trade	The increase in each nation's consumption resulting from specialization in production and trade.
Terms of Trade	The price of one good in terms of the other that two countries agree to trade at. Beneficial terms of trade allow a country to import a good at a lower opportunity cost than the cost for them to produce the good domestically, thus the country gains from trade. In other words, it's the ratio of the price of a nation's export commodity to the price of its import commodity. When more than two commodities are traded, we use the index of export to import prices. Terms of trade are usually given in percentages.
Consumption Frontier	The various alternative combinations of the two commodities that a nation can consume.

Marginal rate of Transformation (MRT)

The amount of one commodity that a nation must give up to get more units of a second commodity. This is another name of the opportunity cost of a commodity. To be more precise, marginal rate of transformation shows the number of goods which will be given up by a country to produce an additional unit of other goods while keeping the factors of production constant.

Production Possibilities Curve

It is a graph which shows various combinations of output that a country can produce by applying its best available resources and technology. In other words, A curve that shows the various alternative combinations of the two commodities that a nation can produce by completely utilising all its factors of production with the best available technology. This curve is also known as the transformation curve or the production frontier.

Opportunity Cost Theory

This opportunity cost theory states that the cost of one commodity is the amount of another commodity that must be given up by a country to release just enough factors of production or resources to enable the production of one additional unit of the first commodity. To be more precise, if a country can produce either commodity A or B, the opportunity cost of commodity A is the amount of the other commodity B that must be given up to get one additional unit of commodity A.

Constant Opportunity Costs

It occurs when the opportunity cost stays the same as a country increase its production of one good. This shows that the resources are easily accommodated or changed from the production of one good to the production of another good. To be more precise, constant opportunity costs theory occurs when the equal amount of one commodity that a nation give up releasing just enough resources to produce each additional unit of another commodity.

Increasing Opportunity Costs

It describes how opportunity costs for a country increase as resources are applied. According to the law of increasing opportunity costs, as one good is produced, the opportunity cost to

produce another good will increase. To be more precise, increasing opportunity costs theory implies that a country by increasing amount of one commodity country must give up to release just enough resources to produce additional unit of another commodity.

Offer Curve

A curve that shows how much a nation requires to import commodity in exchange for various quantities of its export commodity. This curve is also known as the reciprocal demand curve.

2.12 SOME USEFUL REFERENCES

James Ellsworth P.T., *International Economy*, Macmillan Library Reference; 6th Revised edition

Harrod, R.F., *International Economics*, Pitman Pub. Corp; Revised edition

James Gerber. 2014 *International Economics*, 6th Ed., Pearson Education, USA

Krugman, Paul R. and Maurice Obstfeld, 1997. *International Economics: Theory and Policy*, 4th Ed., Addison-Wesley, Massachusetts, USA.

Khan, Shabbir, K., *Gains from International Trade*, Bombay: Asia Publishing House.

Malthus T.R., *Principles of Political Economy*

Ohlin, *Interregional and International Trade*

Sodersten, B. 1980. *International Economics*, Macmillan London

Sodersten, B. and Reed, G. 1994. *International Economics*, 3rd Ed., Macmillan London

Salvatore, D. John 2004. *International Economics*, 8th Ed., Wiley and Sons, Inc. New Jersey

Taussig, F.W., *International Trade*

Viner, Jacob, *Studies in the Theory of International Trades*, London: George Allen and Unwin Ltd.

2.13 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Refer to section 2.2
- 2) Refer to section 2.3
- 3) Refer to section 2.4

Check Your Progress 2

- 1) Refer to section 2.5
- 2) Refer to section 2.5
- 3) Refer to section 2.5

Check Your Progress 3

- 1) Refer to section 2.6
- 2) Refer to section 2.9
- 3) Refer to section 2.6
- 4) Refer to section 2.8

